Label Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Label Quality

> Prepare Label Quality: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Write an observable **label rule** with boundaries and a review option

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a task from the meaning of its target

Reference example, not the only acceptable wording: Route type is classification because 1 and 2 encode categories. Whole-minute duration is regression because numerical differences measure time.

Exceeding 10 minutes is binary classification because the target is yes or no.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

Concept to explain

Ambiguous definitions and inconsistent **annotation** place an upper bound on learnable performance.

Term	Meaning
annotation	human assignment of a label
label rule	observable criterion for assigning a class
agreement	fraction of labels that match
adjudication	resolve disagreements using evidence

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions using invented scene descriptions. The target is applying a stated **observation** rule, not certifying real robot safety. Keep original independent labels separate from reviewed decisions.

Exact **agreement** includes review in the main exercise; review is a workflow status, not a physical lane class. Avoid calling **annotator agreement** accuracy.

Readiness answer

Unknown means the measurement is unavailable. Zero is a recorded value permitted by that schema; replacing unknown with zero invents **evidence**.

Printed lesson and workbook

Scene cards for the four worked examples and five independent cases

Optional Python 3 editor with no extra packages

Blank decision log with item originals **evidence** rule decision reviewer

End goal for the pictorial case

By the end, I can explain label quality using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Label Quality using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Measure **annotator agreement** and revise the labeling guide.

MODEL IDEA Ambiguous definitions and inconsistent **annotation** place an upper bound on learnable performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Two **annotator**s label 20 leaves and agree on 16. Observed **agreement** = 16 ÷ 20 = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Two **annotator**s label 20 leaves and agree on 16.

Step 2

STEP 2 Observed **agreement** = 16 ÷ 20 = **80%**.

Step 3

STEP 3 Most dis**agreement**s concern tiny spots near the boundary.

Step 4

STEP 4 Define a minimum spot rule, relabel a fixed subset, and report the new **agreement**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

The teaching case

Improve label quality.

Exact explanation to model

Why the result follows

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Explain the added worked example

Explain the added worked example · 1

A model learns from examples with recorded targets. If two people interpret a target differently, their labels can mix different tasks.

This week you will write a **label rule**, compare independent labels and keep a record of disputed cases. We use invented corridor descriptions, not real images or a system for controlling a robot.

Explain the added worked example · 2

The classroom task is to label whether a marked practice lane is blocked in a supplied description.

Rule v1 says blocked means an object touches or crosses either lane boundary or lies inside the lane; clear means all objects are outside both boundaries and the entire lane is visible.

Use review if visibility or position is insufficient to decide. Review is a workflow status for a person to inspect, not a third physical condition of the lane.

These are exercise definitions, not real robot safety rules. Do not infer that a lane is physically safe from this label. The label describes only the supplied **evidence** under this version of the rule.

Two **annotator**s used an earlier vague instruction, mark an obstructed lane, on four invented items. Their labels are shown in the code. A and D match; B and C differ.

Compare the same item IDs, not merely the first row in two differently ordered files.

Explain the added worked example · 3

The **output** is 2 4: two matching labels among four paired items, so observed exact **agreement** is 2/4 or 50 percent. Here review is included as a label value in the comparison: review versus clear is a dis**agreement**.

This simple count is not model accuracy and does not establish which labels are correct.

The code assumes one valid pair per unique item ID and allowed label strings. It does not validate IDs or spelling. Use a nonempty paired set before calculating a fraction; zero paired items gives no **agreement** estimate.

Key terms

Term	Meaning
Label rule	A written definition for assigning an outcome to an example
Annotator	A person who assigns labels using a stated rule
Agreement	How often annotators assign the same label to the same items
Adjudication	Reviewing evidence and rules to resolve a disputed label or retain uncertainty

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

For B, the full lane is visible and a wheel touches its boundary. One **annotator** treated touching as clear; the other treated it as blocked. Rule v1 explicitly includes touching, so the reviewed label is blocked.

Preserve both original labels and record the item, rule version, **evidence**, decision and reviewer. Do not silently overwrite the originals.

For C, a curtain hides part of the lane. One **annotator** guessed clear; the other requested review. Rule v1 requires full visibility for clear. The justified status remains review until sufficient **evidence** is available.

More votes cannot reveal the hidden area.

> **Using rule v1, explain the decisions for B and C. If you exclude C from the original comparison, what is agreement among the remaining pairs? Why must you still report the exclusion?**

A decision log could say: B, original clear/blocked, wheel touches line, v1, blocked after review by the teacher. For C it could say: original review/clear, lane partly hidden, v1, unresolved review.

Add the labeling method and rule version to the dataset card.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Label Quality

Write a **label rule** that resolves the visible boundary case.

The facts we will use

> **Supplied facts**
>
> Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/6

Label Quality

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Apply visible rule

X is spotted; Y is healthy under this exercise's definition.



2. Handle uncertainty

Z requires review; hidden tissue cannot be treated as visibly spot-free.



3. Review disagreement

If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

CHANGE ONE THING

Z receives a new fully visible image with no brown spots.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Apply visible rule	X is spotted; Y is healthy under this exercise's definition.
2. Handle uncertainty	Z requires review; hidden tissue cannot be treated as visibly spot-free.
3. Review disagreement	If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

Change one thing in the pictured case

Change one thing

Change: Z receives a new fully visible image with no brown spots.

Predicted result

Prediction: Label the new **observation** healthy under the toy rule.

Why the result follows

Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Label Quality, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Two **annotator**s disagree on three of ten leaves, especially mild spotting.

2. State the decision to explain

Write the question to explain: Improve label quality.

3. Reveal the worked reasoning

Reveal after **prediction**s: Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Hold back the independent task

Keep for an independent attempt: Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better **process**.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: B is blocked because touching counts; C stays review because the lane is partly hidden. Excluding C leaves the original pairs A, B and D: 2 matches out of 3, about 66.7 percent.

Report the exclusion because this is a different subset from the four-item comparison, not newly correct labels. Exit: perfect **agreement** establishes matching labels on these items only. Both people can share a rule error.

Check their labels against the boundary **evidence** and v1 before making a correctness claim.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve unknown versus zero. Ask why a label needs an explicit meaning before it becomes training data.

Stage	Teacher action
7 to 18 minutes	Read v1. Apply inside, boundary, outside and hidden examples. Establish that these are fictional descriptions, not robot safety advice.
18 to 32 minutes	Have pairs label fresh verbal examples independently before talking. Trace the supplied four-item code and complete W1 and W2.
32 to 45 minutes	Compare B and C using evidence. Complete the guided denominator question and preserve the original pair list.
Session 2 start to 8 minutes	Retrieve agreement versus correctness. Show a shared mistake on a touching-boundary example.
8 to 20 minutes	Model a decision-log entry without erasing original labels. Complete W3 and discuss a genuine rule revision.
20 to 35 minutes	Collect W4 independently before support. Require both denominators, rule-based decisions and the unresolved review case.
35 to 45 minutes	Use W5 and the exit response. After feedback, offer the fresh retry and record any support used.

Reduce help deliberately

Completion practice

Completion task: Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

Reasoning to listen for

Teacher check: Two of four matched items have identical **annotator** labels: 2/4 or 50 percent. Review is included in this exact comparison.

Neither **annotator** is established as a correct reference, and no model **prediction**s are being scored.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Label definitions and dis**agreement**. Proposed explanation: "**Agreement** is accuracy". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: **Agreement** compares **annotator**s. Accuracy would require **prediction**s and an appropriate reference; shared mistakes can still match.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Agreement or a majority cannot reveal hidden tissue. Seek an adequate **observation** and preserve the review status until then. Changed-case check: Label the new **observation** healthy under the toy rule.

The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Changed-case reasoning: Label the new **observation** healthy under the toy rule. The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Label Quality investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Label Lab: independently label twenty cards, compute **agreement**, cluster dis**agreement**s, revise one rule, and repeat.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR The model can learn the correct concept even when people label the same example differently.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Agreement is accuracy: **Agreement** compares **annotator**s. Accuracy would require **prediction**s and an appropriate reference; shared mistakes can still match.

Review means clear: Review preserves insufficient **evidence**. It is neither a negative label nor proof of a physical condition.

Voting removes uncertainty: Votes may select a decision policy but cannot create missing **evidence**. Preserve unresolved cases and document the policy.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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Assess and explain the answers

> Use the Label Quality questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and

document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Independent question

Independent question · withhold the key

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better **process**.

Expected independent answer

Teacher key · Expected independent answer

Review the labeling rule and **evidence**, resolve or mark uncertainty, and document the decision independently of model performance.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The most useful label is whichever makes evaluation accuracy highest.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Agreement=80%. Shared misunderstanding can produce **agreement** on wrong labels. Clarify spotting criteria, inspect disputed and agreed examples, use appropriate **adjudication** and record ambiguity.

Do not treat majority **agreement** alone as ground truth.

Explanations for the additional workbook examples

W1 Use the boundary and review rules

Workbook question · W1

Under rule v1, label a wheel touching the boundary, an object entirely outside a fully visible lane, and a partly hidden lane. Explain the **evidence** for each.

Teacher reasoning · W1

Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible.

A partly hidden lane requires review because clear cannot be established from the available description.

W2 Interpret agreement

Workbook question · W2

Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

Teacher reasoning · W2

Two of four matched items have identical **annotator** labels: 2/4 or 50 percent. Review is included in this exact comparison. Neither **annotator** is established as a correct reference, and no model **prediction**s are being scored.

W3 Write a decision log

Workbook question · W3

For B and C, record original labels, visible **evidence**, rule version and reviewed decision. Explain why the original labels should remain available.

Teacher reasoning · W3

B: clear/blocked, wheel touches boundary, v1, blocked. C: review/clear, partly hidden lane, v1, unresolved review. Record the reviewer as well.

Originals preserve the independent dis**agreement** and let another person audit how a decision was made.

W4 Apply the process independently

Workbook question · W4

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

Teacher reasoning · W4

Three of five pairs match, or 60 percent. Reviewed results are P blocked, Q blocked, R review, S clear, T blocked. Excluding R leaves two matches among four pairs, or 50 percent.

The subset omits the uncertain case and does not establish correctness or general reliability. Preserve original labels separately from reviewed decisions.

W5 Draft and test a label rule

Workbook question · W5

Write a short v1 labeling instruction covering blocked, clear, the touching boundary and review. Add one boundary example. Explain why three people voting clear cannot settle a hidden lane.

Teacher reasoning · W5

A suitable instruction specifies the marked lane: inside or touching either boundary means blocked; clear requires all objects outside and full visibility; insufficient position or visibility means review.

A wheel touching the line is blocked. Votes supply no missing visual **evidence**, so a hidden lane remains review under v1.

Assess independent transfer

Independent question · withhold the key

Selected case: New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

Selected case answer

Teacher key · Selected case answer

Three of five pairs match, or 60 percent. Reviewed results are P blocked, Q blocked, R review, S clear, T blocked. Excluding R leaves two matches among four pairs, or 50 percent.

The subset omits the uncertain case and does not establish correctness or general reliability. Preserve original labels separately from reviewed decisions.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Label Quality to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 7: Sampling and Intended Use. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use lane drawings and movable object cards for touching versus outside; cover part of the drawing for review. Provide blank log headings without decisions.

Extension: construct two **annotator**s who agree perfectly while both break v1 and explain why the **agreement** fraction does not reveal the error.

Require W4 counts 3/5 including review and 2/4 excluding review, all five v1 decisions, and an explicit subset limitation.

Fresh retry: U clear/clear with all objects outside a fully visible lane; V blocked/clear with an object touching the boundary; W blocked/blocked with an object inside; X review/blocked with part of the lane hidden.

Original **agreement** is 2/4 or 50 percent; excluding X gives 2/3 or about 66.7 percent. Reviewed v1 decisions are clear, blocked, blocked, review. Require **evidence** for V and X before accepting the retry.

Week 6 covers splits and related examples. Carry forward item IDs, rule versions, original labels and review status so that related cases and labeling decisions remain traceable.

Plan the delayed return

Suggested return points in this level: ai-4/7, ai-4/9, ai-4/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	

Record	Teacher evidence
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Label Quality lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Write an observable **label rule** with boundaries and a review option

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

Supported practice answer

Reasoned practice answer: Two of four matched items have identical **annotator** labels: 2/4 or 50 percent. Review is included in this exact comparison.

Neither **annotator** is established as a correct reference, and no model **prediction**s are being scored.

Check every learner before discussion: Use the worked case for Label definitions and dis**agreement**. Proposed explanation: "**Agreement** is accuracy". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Agreement** compares **annotator**s. Accuracy would require **prediction**s and an appropriate reference; shared mistakes can still match.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Z receives a new fully visible image with no brown spots. Require a **prediction** and a reason before comparing. Expected result: Label the new **observation** healthy under the toy rule.

Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

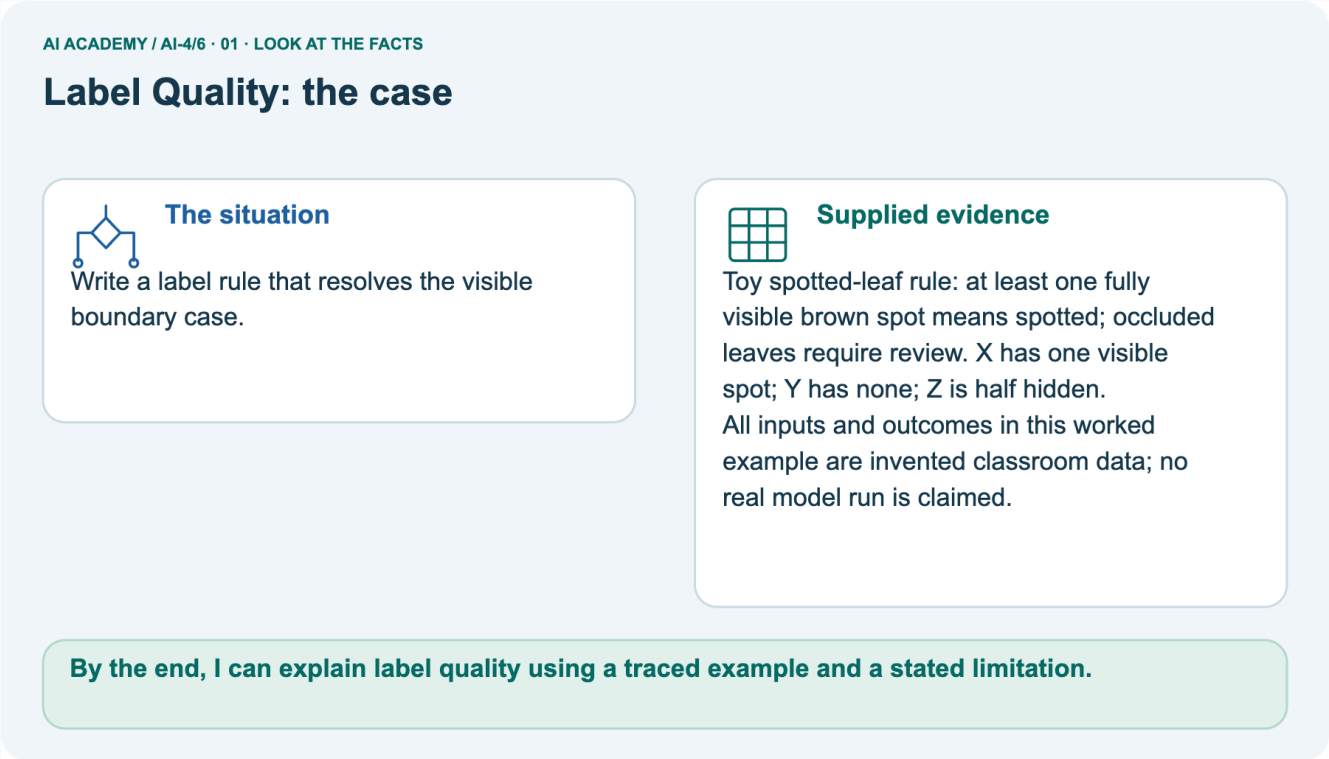
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Label Quality graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Label Quality: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a **label rule** that resolves the visible boundary case. Supplied facts: Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review.

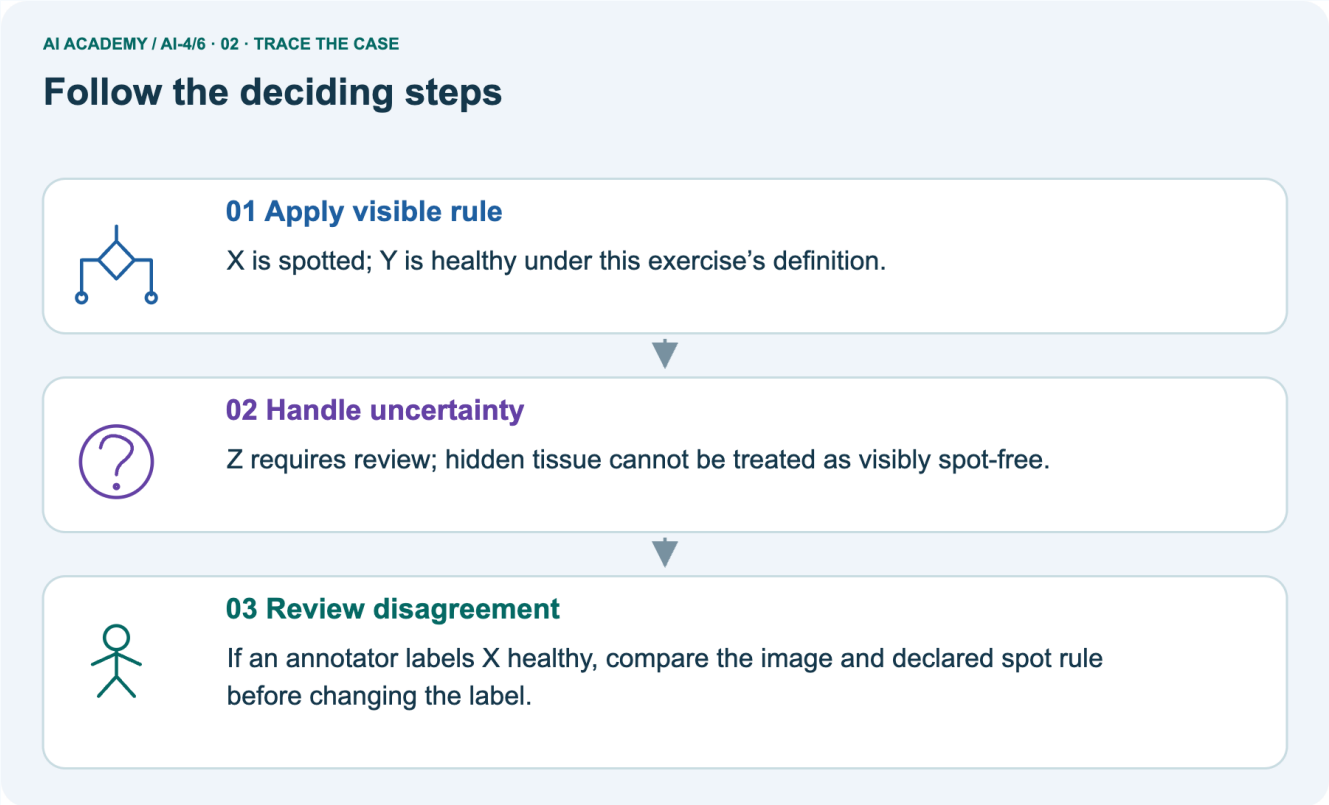
X has one visible spot; Y has none; Z is half hidden. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply visible rule: X is spotted; Y is healthy under this exercise's definition. → Handle uncertainty: Z requires review; hidden tissue cannot be treated as visibly spot-free. → Review dis**agreement**: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow a disagreement back to evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

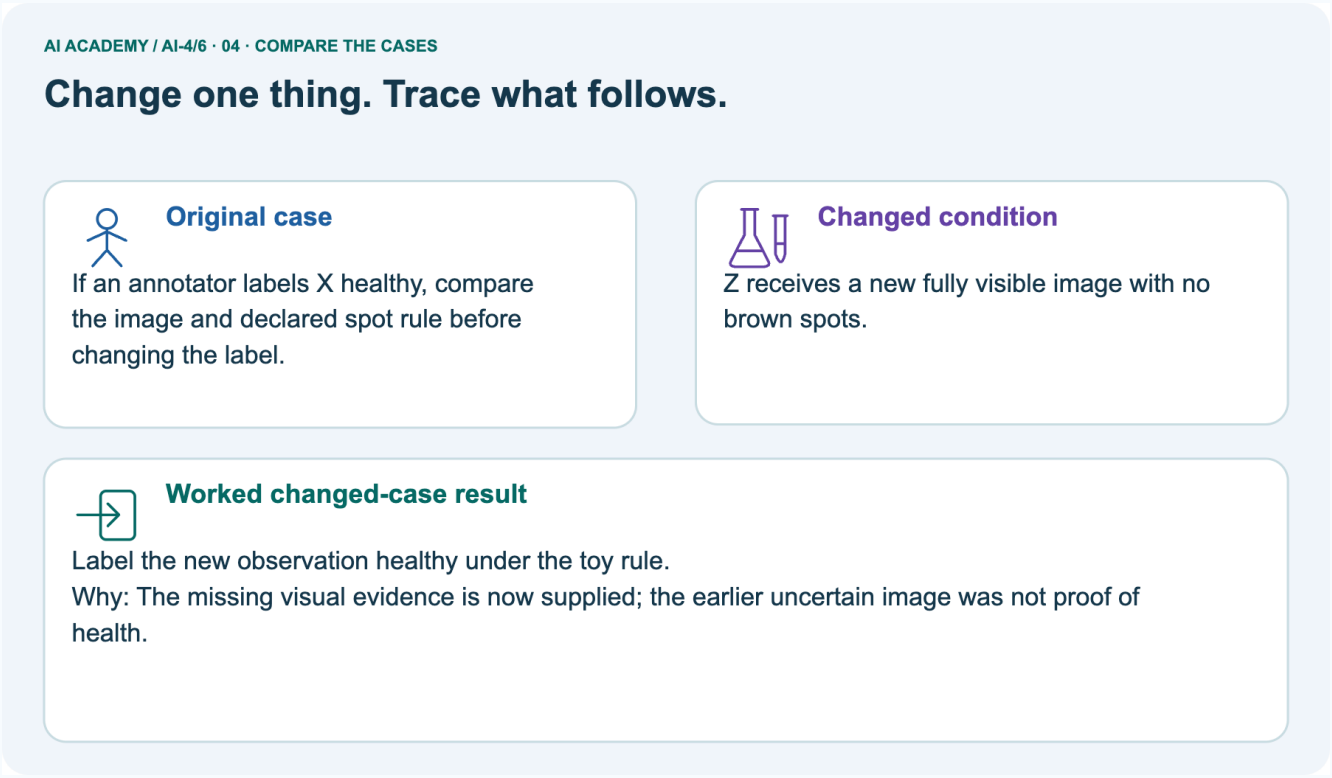
Follow a dis**agreement** back to **evidence**. B has two labels | Read boundary rule | Keep labels and reason. C includes review | Check visible **evidence** | Retain uncertainty if needed.

A and D match | Check rule as well | **Agreement** is not proof

> Read the labels and values. Explain the relationship the model shows.

> **Discuss while showing the picture**
>
> Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Changed condition: Z receives a new fully visible image with no brown spots.

Expected result: Label the new **observation** healthy under the toy rule. Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

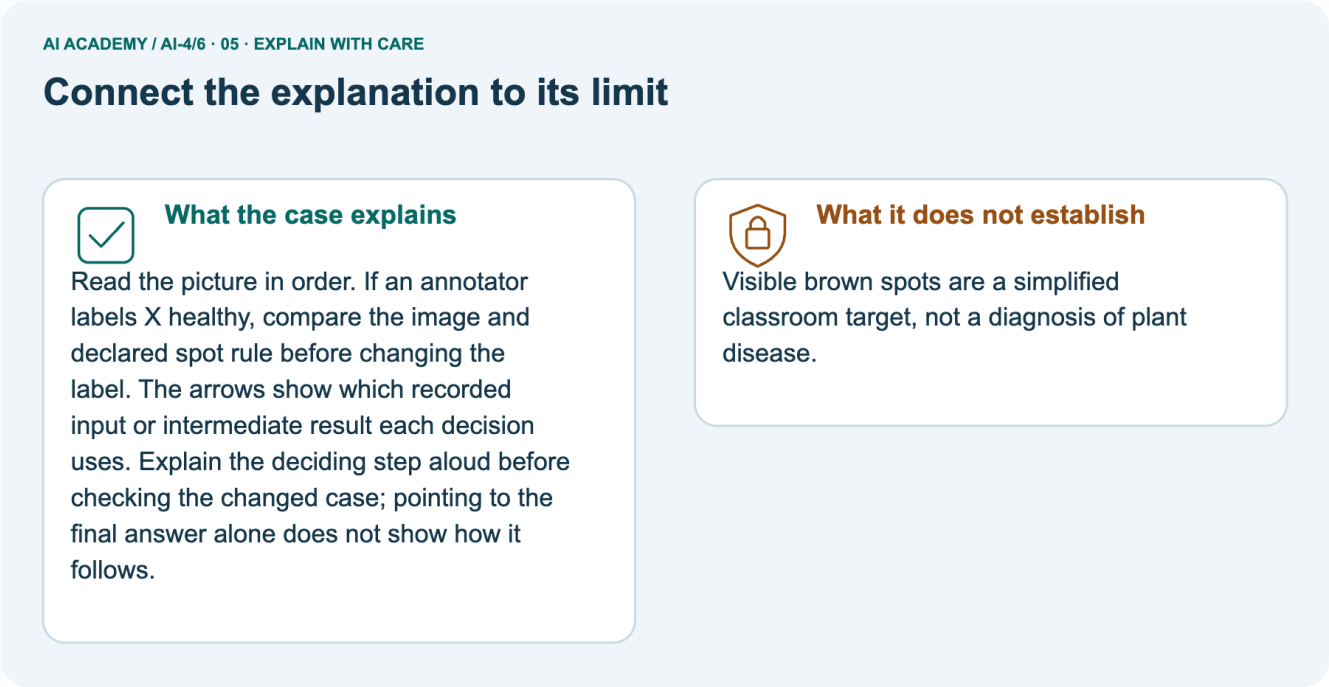
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

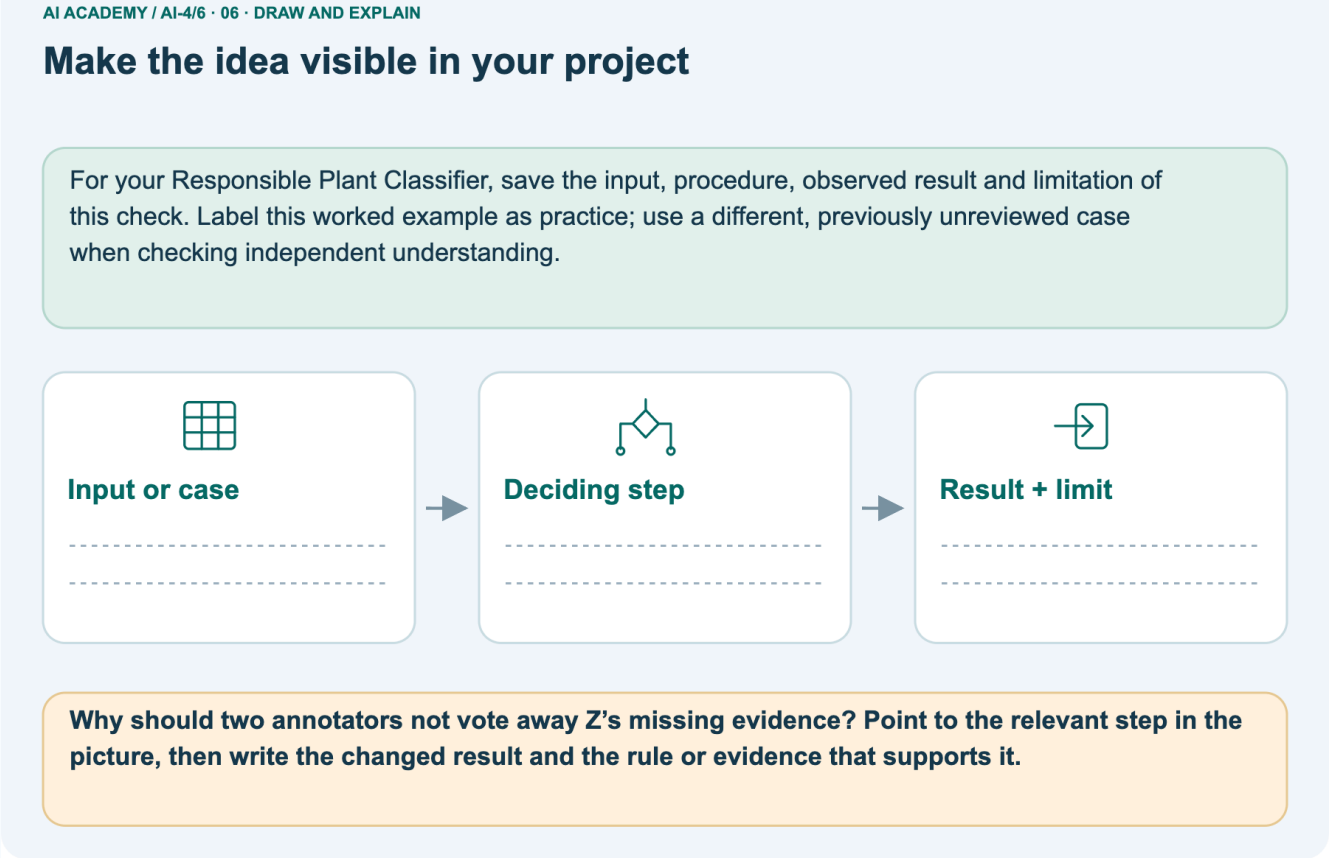
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: **Agreement** or a majority cannot reveal hidden tissue. Seek an adequate **observation** and preserve the review status until then. Changed-case check: Label the new **observation** healthy under the toy rule.

The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.